



Nd:YAG laser

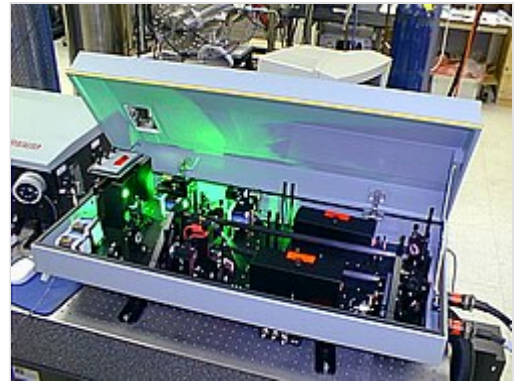
Nd:YAG (**neodymium-doped yttrium aluminum garnet**; **Nd:Y₃Al₅O₁₂**) is a crystal that is used as a lasing medium for solid-state lasers. The dopant, neodymium in the +3 oxidation state, Nd(III), typically replaces a small fraction (1%) of the yttrium ions in the host crystal structure of the yttrium aluminum garnet (YAG), since the two ions are of similar size.^[1] It is the neodymium ion which provides the lasing activity in the crystal, in the same fashion as red chromium ion in ruby lasers.^[1]

Laser operation of Nd:YAG was first demonstrated by Joseph E. Geusic et al. at Bell Laboratories in 1964.^[2] Geusic and LeGrand Van Uitert received the Optical Society of America's R. W. Wood Prize in 1993 “for the discovery of the Nd:YAG laser and the demonstration of its usefulness as a practical solid state laser source”.

Technology

Nd:YAG lasers are optically pumped using a flashtube or laser diodes. These are one of the most common types of laser, and are used for many different applications. Nd:YAG lasers typically emit light with a wavelength of 1064 nm, in the infrared.^[3] However, there are also transitions near 946, 1120, 1320, and 1440 nm. Nd:YAG lasers operate in both pulsed and continuous mode. Pulsed Nd:YAG lasers are typically operated in the so-called Q-switching mode: An optical switch is inserted in the laser cavity waiting for a maximum population inversion in the neodymium ions before it opens. Then the light wave can run through the cavity, depopulating the excited laser medium at maximum population inversion. In this Q-switched mode, output powers of 250 megawatts and pulse durations of 10 to 25 nanoseconds have been achieved.^[4] The high-intensity pulses may be efficiently frequency doubled to generate laser light at 532 nm, or higher harmonics at 355, 266 and 213 nm.

Nd:YAG absorbs mostly in the bands between 730–760 nm and 790–820 nm.^[3] At low current densities krypton flashlamps have higher output in those bands than do the more common xenon lamps, which produce more light at around 900 nm. The former are therefore more efficient for pumping Nd:YAG lasers.^[5]



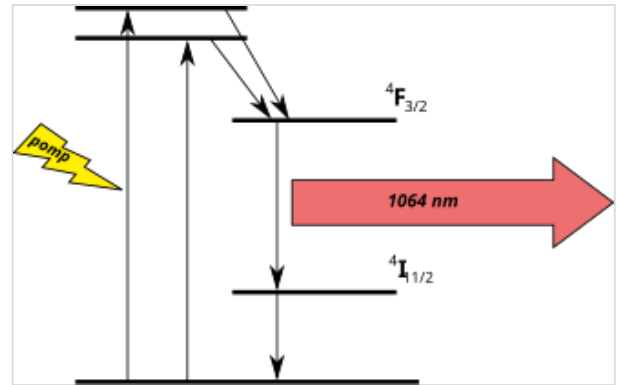
Nd:YAG laser with lid open showing frequency-doubled 532 nm green light



Nd:YAG laser rod

The amount of the neodymium dopant in the material varies according to its use. For continuous wave output, the doping is significantly lower than for pulsed lasers. The lightly doped CW rods can be optically distinguished by being less colored, almost white, while higher-doped rods are pink-purple.

Other common host materials for neodymium are: YLF (yttrium lithium fluoride, 1047 and 1053 nm), YVO₄ (yttrium orthovanadate, 1064 nm), and glass. A particular host material is chosen in order to obtain a desired combination of optical, mechanical, and thermal properties. Nd:YAG lasers and variants are pumped either by flashtubes, continuous gas discharge lamps, or near-infrared laser diodes (DPSS lasers). Prestabilized laser (PSL) types of Nd:YAG lasers have proved to be particularly useful in providing the main beams for gravitational wave interferometers such as LIGO, VIRGO, GEO600 and TAMA.



Neodymium ions in various types of ionic crystals, and also in glasses, act as a laser gain medium, typically emitting 1064 nm light from a particular atomic transition in the neodymium ion, after being "pumped" into excitation from an external source

Applications

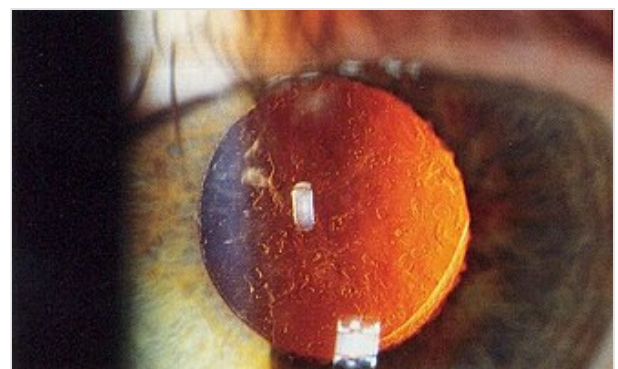
Medicine

Nd:YAG lasers are used in ophthalmology to correct posterior capsular opacification,^[6] after cataract surgery, for peripheral iridotomy in patients with chronic^[7] and acute angle-closure glaucoma,^[8] where it has largely superseded surgical iridectomy,^[9] for the treatment of vitreous eye floaters,^[10] for pan-retinal photocoagulation in the treatment of proliferative diabetic retinopathy,^[11] and to damage the retina in ophthalmology animal research.^[12]

Nd:YAG lasers emitting light at 1064 nm have been the most widely used laser for laser-induced thermotherapy, in which benign or malignant lesions in various organs are ablated by the beam.

In oncology, Nd:YAG lasers can be used to remove skin cancers.^[13] They are also used to reduce benign thyroid nodules,^[14] and to destroy primary and secondary malignant liver lesions.^{[15][16]}

To treat benign prostatic hyperplasia (BPH), Nd:YAG lasers can be used for laser prostate surgery—a form of transurethral resection of the prostate.^{[17][18]}



Slit lamp photograph of posterior capsular opacification visible a few months after implantation of intraocular lens in eye, seen on retroillumination

These lasers are also used extensively in the field of cosmetic medicine for laser hair removal and the treatment of minor vascular defects such as spider veins on the face and legs. Nd:YAG lasers are also used to treat venous lake lip lesions.^[19] Recently Nd:YAG lasers have been used for treating dissecting cellulitis of the scalp, a rare skin disease.^[20]

Using hysteroscopy the Nd:YAG laser has been used for removal of uterine septa within the inside of the uterus.^[21]

In podiatry, the Nd:YAG laser is being used to treat onychomycosis, which is fungus infection of the toenail.^[22] The merits of laser treatment of these infections are not yet clear, and research is being done to establish effectiveness.^{[23][24]}

Dentistry

Nd:YAG dental lasers have been used for the removal of dental caries as an alternative to drill therapy, although evidence supporting its use is of low quality.^[25] They have also been used for soft tissue surgeries in the oral cavity, such as gingivectomy,^{[26][27]} periodontal sulcular debridement,^[28] LANAP,^[29] and pulpotomy.^[30] Nd:YAG dental lasers have also been shown to be effective at treating and preventing dental hypersensitivity,^[31] as an adjunct for periodontal instrumentation,^[32] and for the treatment of recurrent aphthous stomatitis.^[33]

Manufacturing

Nd:YAG lasers are used in manufacturing for engraving, etching, or marking a variety of metals and plastics, or for metal surface enhancement processes like laser peening.^[34] They are extensively used in manufacturing for cutting and welding steel, semiconductors and various alloys. For automotive applications (cutting and welding steel) the power levels are typically 1–5 kW. Super alloy drilling (for gas turbine parts) typically uses pulsed Nd:YAG lasers (millisecond pulses, not Q-switched). Nd:YAG lasers are also employed to make subsurface markings in transparent materials such as glass or acrylic glass and in white and transparent polycarbonate for identity documents. Lasers of up to 2 kW are used for selective laser melting of metals in additive layered manufacturing. In aerospace applications, they can be used to drill cooling holes for enhanced air flow/heat exhaust efficiency.

Nd:YAG lasers are also used in the non-conventional rapid prototyping process laser engineered net shaping (LENS).

Laser peening typically uses a high energy (10 to 40 joule) 10 to 30 nanosecond pulse. The laser beam is focused down to a few millimeters in diameter to deposit gigawatts of power on the surface of a part. Laser peening is unlike other manufacturing processes in that it neither heats nor adds material; it is a mechanical process of cold working the metallic component to impart compressive residual stresses. Laser peening is widely used in gas-fired turbine engines in both aerospace and power generation to increase strength and improve resistance to damage and metal fatigue.^[35]

Fluid dynamics

Nd:YAG lasers can be used for flow visualization techniques in fluid dynamics (for example particle image velocimetry or laser-induced fluorescence).^[36]

Biophysics

Nd:YAG lasers are frequently used to build optical tweezers for biological applications. This is because Nd:YAG lasers mostly emit at a wavelength of 1064 nm. Biological samples have a low absorption coefficient at this wavelength, as biological samples are usually mostly made up of water. ^[37] As such, using an Nd:YAG laser minimizes the damage to the biological sample being studied.

Automotive

Researchers from Japan's National Institutes of Natural Sciences are developing laser igniters that use YAG chips to ignite fuel in an engine, in place of a spark plug.^{[38][39]} The lasers use several 800 picosecond long pulses to ignite the fuel, producing faster and more uniform ignition. The researchers say that such igniters could yield better performance and fuel economy, with fewer harmful emissions.

Military

The Nd:YAG laser is the most common laser used in laser designators and laser rangefinders.

During the Iran–Iraq War, Iranian soldiers suffered more than 4000 cases of laser eye injury, caused by a variety of Iraqi sources including tank rangefinders. The 1064 nm wavelength of Nd:YAG is thought to be particularly dangerous, as it is invisible and initial exposure is painless.^[40]

The Chinese ZM-87 blinding laser weapon uses a laser of this type, though only 22 have been produced due to their prohibition by the Convention on Certain Conventional Weapons. North Korea is reported to have used one of these weapons against American helicopters in 2003.^{[41][42]}



Military surplus Nd:YAG laser rangefinder firing. The laser fires through a collimator, focusing the beam, which blasts a hole through a rubber block, releasing a burst of plasma.

Cavity ring-down spectroscopy (CRDS)

The Nd:YAG may be used in the application of cavity ring-down spectroscopy, which is used to measure the concentration of some light-absorbing substance.^[43]

Laser-induced breakdown spectroscopy (LIBS)

A range of Nd:YAG lasers are used in analysis of elements in the periodic table. Though the application by itself is fairly new with respect to conventional methods such as XRF or ICP, it has proven to be less time consuming and a cheaper option to test element concentrations. A high-power Nd:YAG laser is focused onto the sample surface to produce plasma. Light from the plasma is captured by spectrometers and the characteristic spectra of each element can be identified, allowing concentrations of elements in the sample to be measured.

Laser pumping

Nd:YAG lasers, mainly via their second and third harmonics, are widely used to excite dye lasers either in the liquid^[44] or solid state.^[45] They are also used as pump sources for vibronically broadened solid-state lasers such as Cr⁴⁺:YAG or via the second harmonic for pumping Ti:sapphire lasers.

Additional frequencies

For many applications, the infrared light is frequency-doubled or -tripled using nonlinear optical materials such as lithium triborate to obtain visible (532 nm, green) or ultraviolet light.^[46] Cesium lithium borate generates the 4th and 5th harmonics of the Nd:YAG 1064 nm fundamental wavelength.^[47] A green laser pointer is a frequency doubled Nd:YVO₄ diode-pumped solid state laser (DPSS laser).^[48] Nd:YAG can be also made to lase at its non-principal wavelength. The line at 946 nm is typically employed in "blue laser pointer" DPSS lasers, where it is doubled to 473 nm.^{[49][50][51]}

Physical and chemical properties of Nd:YAG

Properties of YAG crystal

- Formula: Y₃Al₅O₁₂
- Molecular weight: 596.7
- Crystal structure: Cubic
- Hardness: 8–8.5 (Mohs)^[52]
- Melting point: 1970 °C (3540 °F)
- Density: 4.55 g/cm³

Refractive index of Nd:YAG

Wavelength (μm)	Index <i>n</i> (25 °C)
0.8	1.8245
0.9	1.8222
1.0	1.8197
1.2	1.8152
1.4	1.8121
1.5	1.8121

Properties of Nd:YAG @ 25 °C (with 1% Nd doping)

- Formula: Y_{2.97}Nd_{0.03}Al₅O₁₂
- Weight of Nd: 0.725%
- Atoms of Nd per unit volume: 1.38×10²⁰ /cm³

- Charge state of Nd: 3^+
- Emission wavelength: 1064 nm
- Transition: $^4F_{3/2} \rightarrow ^4I_{11/2}$
- Duration of fluorescence: 230 μs ^[52]
- Thermal conductivity: $0.14 \text{ W}\cdot\text{cm}^{-1}\cdot\text{K}^{-1}$
- Specific heat capacity: $0.59 \text{ J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$
- Thermal expansion: $6.9\times 10^{-6} \text{ K}^{-1}$
- dn/dT : $7.3\times 10^{-6} \text{ K}^{-1}$
- Young's modulus: $3.17\times 10^4 \text{ K}\cdot\text{g}/\text{mm}^{-2}$
- Poisson's ratio: 0.25
- Resistance to thermal shock: $790 \text{ W}\cdot\text{m}^{-1}$

References and notes

1. Koechner §2.3, pp. 48–53.
2. Geusic, J. E.; Marcos, H. M.; Van Uitert, L. G. (1964). "Laser oscillations in nd-doped yttrium aluminum, yttrium gallium and gadolinium garnets". *Applied Physics Letters*. **4** (10): 182. Bibcode:1964ApPhL...4..182G (<https://ui.adsabs.harvard.edu/abs/1964ApPhL...4..182G>). doi:10.1063/1.1753928 (<https://doi.org/10.1063%2F1.1753928>).
3. Yariv, Amnon (1989). *Quantum Electronics* (3rd ed.). Wiley. pp. 208–11. ISBN 978-0-471-60997-1.
4. Walter Koechner (1965) *Solid-state laser engineering*, Springer-Verlag, p. 507
5. Koechner §6.1.1, pp. 251–64.
6. Findl, Oliver; Buehl, Wolf; Bauer, Peter; Sycha, Thomas (2010-02-17). "Interventions for preventing posterior capsule opacification" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10658648>). *The Cochrane Database of Systematic Reviews* (2): CD003738. doi:10.1002/14651858.CD003738.pub3 (<https://doi.org/10.1002%2F14651858.CD003738.pub3>). ISSN 1469-493X (<https://search.worldcat.org/issn/1469-493X>). PMC 10658648 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10658648>). PMID 20166069 (<https://pubmed.ncbi.nlm.nih.gov/20166069>).
7. Dias-Santos, Arnaldo; Ferreira, Joana; Abegão Pinto, Luís; Domingues, Isabel; Silva, José Pedro; Cunha, João Paulo; Reina, Maria (April 2015). "Phacoemulsification versus peripheral iridotomy in the management of chronic primary angle closure: long-term follow-up" (<https://pubmed.ncbi.nlm.nih.gov/24728533/>). *International Ophthalmology*. **35** (2): 173–178. doi:10.1007/s10792-014-9926-8 (<https://doi.org/10.1007%2Fs10792-014-9926-8>). hdl:10400.17/2093 (<https://hdl.handle.net/10400.17%2F2093>). ISSN 1573-2630 (<https://search.worldcat.org/issn/1573-2630>). PMID 24728533 (<https://pubmed.ncbi.nlm.nih.gov/24728533>). S2CID 14929770 (<https://api.semanticscholar.org/CorpusID:14929770>).
8. Saunders, D. C. (September 1990). "Acute closed-angle glaucoma and Nd-YAG laser iridotomy" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1042198>). *The British Journal of Ophthalmology*. **74** (9): 523–525. doi:10.1136/bjo.74.9.523 (<https://doi.org/10.1136%2Fbjo.74.9.523>). ISSN 0007-1161 (<https://search.worldcat.org/issn/0007-1161>). PMC 1042198 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1042198>). PMID 2393642 (<https://pubmed.ncbi.nlm.nih.gov/2393642>).

9. Rivera, A. H.; Brown, R. H.; Anderson, D. R. (September 1985). "Laser iridotomy vs surgical iridectomy. Have the indications changed?" (<https://pubmed.ncbi.nlm.nih.gov/4038128/>). *Archives of Ophthalmology*. **103** (9): 1350–1354. doi:10.1001/archophth.1985.01050090102042 (<https://doi.org/10.1001%2Farchophth.1985.01050090102042>). ISSN 0003-9950 (<https://search.worldcat.org/issn/0003-9950>). PMID 4038128 (<https://pubmed.ncbi.nlm.nih.gov/4038128>).
10. Kokavec J, Wu Z, Sherwin JC, Ang AJ, Ang GS (2017). "Nd:YAG laser vitreolysis versus pars plana vitrectomy for vitreous floaters" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6481890>). *Cochrane Database Syst Rev*. **2017** (6): CD011676. doi:10.1002/14651858.CD011676.pub2 (<https://doi.org/10.1002%2F14651858.CD011676.pub2>). PMC 6481890 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6481890>). PMID 28570745 (<https://pubmed.ncbi.nlm.nih.gov/28570745>).
11. Moutray, Tanya; Evans, Jennifer R.; Lois, Noemi; Armstrong, David J.; Peto, Tunde; Azuara-Blanco, Augusto (2018-03-15). "Different lasers and techniques for proliferative diabetic retinopathy" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6494342>). *The Cochrane Database of Systematic Reviews*. **2018** (3): CD012314. doi:10.1002/14651858.CD012314.pub2 (<https://doi.org/10.1002%2F14651858.CD012314.pub2>). ISSN 1469-493X (<https://search.worldcat.org/issn/1469-493X>). PMC 6494342 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6494342>). PMID 29543992 (<https://pubmed.ncbi.nlm.nih.gov/29543992>).
12. Kameel Ghaly, Sally; Foad Ghoneim, Dina; Abdelkawi Ahmed, Salwa; Medhat Abdel-Salam, Ahmed (2013). "Histological Evaluation of Retina after Photo Disruption for Vitreous Humor by Q-Switched Neodymium-Doped Yttrium Aluminium Garnet (Nd:YAG) Laser" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4282007>). *Journal of Lasers in Medical Sciences*. **4** (4): 190–198. ISSN 2008-9783 (<https://search.worldcat.org/issn/2008-9783>). PMC 4282007 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4282007>). PMID 25606329 (<https://pubmed.ncbi.nlm.nih.gov/25606329>).
13. Moskalik, K; A Kozlov; E Demin; E Boiko (2009). "The Efficacy of Facial Skin Cancer Treatment with High-Energy Pulsed Neodymium and Nd:YAG Lasers". *Photomedicine Laser Surgery*. **27** (2): 345–49. doi:10.1089/pho.2008.2327 (<https://doi.org/10.1089%2Fpho.2008.2327>). PMID 19382838 (<https://pubmed.ncbi.nlm.nih.gov/19382838>).
14. Valcavi R, Riganti F, Bertani A, Formisano D, Pacella CM (November 2010). "Percutaneous Laser Ablation of Cold Benign Thyroid Nodules: A 3-Year Follow-Up Study in 122 Patients". *Thyroid*. **20** (11): 1253–61. doi:10.1089/thy.2010.0189 (<https://doi.org/10.1089%2Fthy.2010.0189>). PMID 20929405 (<https://pubmed.ncbi.nlm.nih.gov/20929405>).
15. Pacella CM; Francica G; Di Lascio FM; Arienti V; Antico E; Caspani B; Magnolfi F; Megna AS; Pretolani S; Regine R; Sponza M; Stasi R (June 2009). "Long-term outcome of cirrhotic patients with early hepatocellular carcinoma treated with ultrasound-guided percutaneous laser ablation: a retrospective analysis" (<https://doi.org/10.1200%2FJCO.2008.19.0082>). *Journal of Clinical Oncology*. **27** (16): 2615–21. doi:10.1200/JCO.2008.19.0082 (<https://doi.org/10.1200%2FJCO.2008.19.0082>). PMID 19332729 (<https://pubmed.ncbi.nlm.nih.gov/19332729>). S2CID 23374952 (<https://api.semanticscholar.org/CorpusID:23374952>).
16. Pompili M; Pacella CM; Francica G; Angelico M; Tisone G; Craboledda P; Nicolardi E; Rapaccini GL; Gasbarrini G . (June 2010). "Percutaneous laser ablation of hepatocellular carcinoma in patients with liver cirrhosis awaiting liver transplantation". *European Journal of Radiology*. **74** (3): e6 – e11. doi:10.1016/j.ejrad.2009.03.012 (<https://doi.org/10.1016%2Fj.ejrad.2009.03.012>). hdl:2108/22095 (<https://hdl.handle.net/2108%2F22095>). PMID 19345541 (<https://pubmed.ncbi.nlm.nih.gov/19345541>).

17. Sun, Feng; Sun, Xincheng; Shi, Qinglu; Zhai, Yuzhang (December 2018). "Transurethral procedures in the treatment of benign prostatic hyperplasia: A systematic review and meta-analysis of effectiveness and complications" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6320039>). *Medicine*. **97** (51): e13360. doi:10.1097/MD.0000000000013360 (<https://doi.org/10.1097%2FMD.0000000000013360>). ISSN 1536-5964 (<https://search.worldcat.org/issn/1536-5964>). PMC 6320039 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6320039>). PMID 30572440 (<https://pubmed.ncbi.nlm.nih.gov/30572440>).
18. Costello, A. J.; Johnson, D. E.; Bolton, D. M. (1992). "Nd:YAG laser ablation of the prostate as a treatment for benign prostatic hypertrophy" (<https://pubmed.ncbi.nlm.nih.gov/1374142/>). *Lasers in Surgery and Medicine*. **12** (2): 121–124. doi:10.1002/lsm.1900120202 (<https://doi.org/10.1002%2Flsm.1900120202>). ISSN 0196-8092 (<https://search.worldcat.org/issn/0196-8092>). PMID 1374142 (<https://pubmed.ncbi.nlm.nih.gov/1374142>). S2CID 39538383 (<https://api.semanticscholar.org/CorpusID:39538383>).
19. Azevedo, L. H; Galletta, V. C; De Paula Eduardo, C; Migliari, D. A (2010). "Venous Lake of the Lips Treated Using Photocoagulation with High-Intensity Diode Laser" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2957073>). *Photomedicine and Laser Surgery*. **28** (2): 263–265. doi:10.1089/pho.2009.2564 (<https://doi.org/10.1089%2Fpho.2009.2564>). PMC 2957073 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2957073>). PMID 19811083 (<https://pubmed.ncbi.nlm.nih.gov/19811083>).
20. Krasner BD; Hamzavi FH; Murakawa GJ; Hamzavi IH (August 2006). "Dissecting cellulitis treated with the long-pulsed Nd:YAG laser". *Dermatologic Surgery*. **32** (8): 1039–44. doi:10.1111/j.1524-4725.2006.32227.x (<https://doi.org/10.1111%2Fj.1524-4725.2006.32227.x>). PMID 16918566 (<https://pubmed.ncbi.nlm.nih.gov/16918566>). S2CID 31317584 (<https://api.semanticscholar.org/CorpusID:31317584>).
21. Yang J, Yin TL, Xu WM, Xia LB, Li AB, Hu J (2006). "Reproductive outcome of septate uterus after hysteroscopic treatment with neodymium:YAG laser". *Photomedicine Laser Surgery*. **24** (5): 625. doi:10.1089/pho.2006.24.625 (<https://doi.org/10.1089%2Fpho.2006.24.625>). PMID 17069494 (<https://pubmed.ncbi.nlm.nih.gov/17069494>).
22. Ledon, Jennifer A.; Savas, Jessica; Franca, Katlein; Chacon, Anna; Nouri, Keyvan (2012). "Laser and light therapy for onychomycosis: a systematic review". *Lasers in Medical Science*. **29** (2): 823–29. doi:10.1007/s10103-012-1232-y (<https://doi.org/10.1007%2Fs10103-012-1232-y>). ISSN 0268-8921 (<https://search.worldcat.org/issn/0268-8921>). PMID 23179307 (<https://pubmed.ncbi.nlm.nih.gov/23179307>). S2CID 7950300 (<https://api.semanticscholar.org/CorpusID:7950300>).
23. Mozena, John; Haverstock, Brent (May 2010). "Laser care for onychomycosis: can it be effective?" (<http://www.podiatrytoday.com/laser-care-for-onychomycosis-can-it-be-effective>). *Podiatry Today*. **23** (5): 54–59.
24. Mozena, John D.; Mitnick, Joshua P. (October 2009). "Emerging concepts in treating onychomycosis" (<http://www.podiatrytoday.com/emerging-concepts-in-treating-onychomycosis>). *Podiatry Today*. **22** (10): 46–51.
25. Montedori, Alessandro; Abraha, Iosief; Orso, Massimiliano; D'Errico, Potito Giuseppe; Pagano, Stefano; Lombardo, Guido (2016-09-26). "Lasers for caries removal in deciduous and permanent teeth" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6457657>). *The Cochrane Database of Systematic Reviews*. **2016** (9): CD010229. doi:10.1002/14651858.CD010229.pub2 (<https://doi.org/10.1002%2F14651858.CD010229.pub2>). ISSN 1469-493X (<https://search.worldcat.org/issn/1469-493X>). PMC 6457657 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6457657>). PMID 27666123 (<https://pubmed.ncbi.nlm.nih.gov/27666123>).
26. Cortes, M. (April 1999). "Nd:YAG laser-assisted gingivectomy, bleaching, and porcelain laminates, Part 2" (<https://pubmed.ncbi.nlm.nih.gov/10765801/>). *Dentistry Today*. **18** (4): 52–55. ISSN 8750-2186 (<https://search.worldcat.org/issn/8750-2186>). PMID 10765801 (<https://pubmed.ncbi.nlm.nih.gov/10765801>).

27. De Benedittis, Michele; Petruzzi, Massimo; Pastore, Luca; Inchingolo, Francesco; Serpico, Rosario (February 2007). "Nd:YAG laser for gingivectomy in Sturge-Weber syndrome" (<https://pubmed.ncbi.nlm.nih.gov/17236940/>). *Journal of Oral and Maxillofacial Surgery*. **65** (2): 314–316. doi:10.1016/j.joms.2006.05.011 (<https://doi.org/10.1016%2Fj.joms.2006.05.011>). ISSN 0278-2391 (<https://search.worldcat.org/issn/0278-2391>). PMID 17236940 (<https://pubmed.ncbi.nlm.nih.gov/17236940/>).
28. David M. Harris, Robert H. Gregg, Delwin K. McCarthy, Leigh E. Colby, Lloyd V. Tilt, "Sulcular debridement with pulsed Nd:YAG," Proc. SPIE 4610, Lasers in Dentistry VIII, (3 June 2002); doi: 10.1117/12.469328
29. Yukna, Raymond A.; Carr, Ronald L.; Evans, Gerald H. (December 2007). "Histologic evaluation of an Nd:YAG laser-assisted new attachment procedure in humans" (<https://pubmed.ncbi.nlm.nih.gov/18092452/>). *The International Journal of Periodontics & Restorative Dentistry*. **27** (6): 577–587. ISSN 0198-7569 (<https://search.worldcat.org/issn/0198-7569>). PMID 18092452 (<https://pubmed.ncbi.nlm.nih.gov/18092452/>).
30. De Coster, Peter; Rajasekharan, Sivaprakash; Martens, Luc (November 2013). "Laser-assisted pulpotomy in primary teeth: a systematic review" (<https://pubmed.ncbi.nlm.nih.gov/23171469/>). *International Journal of Paediatric Dentistry*. **23** (6): 389–399. doi:10.1111/ipd.12014 (<https://doi.org/10.1111%2Fipd.12014>). ISSN 1365-263X (<https://search.worldcat.org/issn/1365-263X>). PMID 23171469 (<https://pubmed.ncbi.nlm.nih.gov/23171469/>).
31. Rezazadeh, Fahimeh; Dehghanian, Paria; Jafarpour, Dana (2019). "Laser Effects on the Prevention and Treatment of Dentinal Hypersensitivity: A Systematic Review" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6499583>). *Journal of Lasers in Medical Sciences*. **10** (1): 1–11. doi:10.15171/jlms.2019.01 (<https://doi.org/10.15171%2Fjlms.2019.01>). ISSN 2008-9783 (<https://search.worldcat.org/issn/2008-9783>). PMC 6499583 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6499583>). PMID 31360362 (<https://pubmed.ncbi.nlm.nih.gov/31360362/>).
32. Roncati, Marisa; Gariffo, Annalisa (April 2014). "Systematic review of the adjunctive use of diode and Nd:YAG lasers for nonsurgical periodontal instrumentation" (<https://pubmed.ncbi.nlm.nih.gov/24697584/>). *Photomedicine and Laser Surgery*. **32** (4): 186–197. doi:10.1089/pho.2013.3695 (<https://doi.org/10.1089%2Fpho.2013.3695>). ISSN 1557-8550 (<https://search.worldcat.org/issn/1557-8550>). PMID 24697584 (<https://pubmed.ncbi.nlm.nih.gov/24697584/>).
33. Suter, Valerie G. A.; Sjölund, Sophia; Bornstein, Michael M. (May 2017). "Effect of laser on pain relief and wound healing of recurrent aphthous stomatitis: a systematic review" (<https://pubmed.ncbi.nlm.nih.gov/28345122/>). *Lasers in Medical Science*. **32** (4): 953–963. doi:10.1007/s10103-017-2184-z (<https://doi.org/10.1007%2Fs10103-017-2184-z>). ISSN 1435-604X (<https://search.worldcat.org/issn/1435-604X>). PMID 28345122 (<https://pubmed.ncbi.nlm.nih.gov/28345122/>). S2CID 3853214 (<https://api.semanticscholar.org/CorpusID:3853214>).
34. Studies on laser peening of spring steel for automotive applications | Ranganathan Kandasamy - Academia.edu (https://www.academia.edu/6125762/Studies_on_laser_peening_of_spring_steel_for_automotive_applications)
35. "How laser peening works" (<http://www.lsptechnologies.com/why-laser-peen.php>). *LSP Technologies*. Retrieved 2022-12-24.
36. Palafox, Gilbert N.; Wicker, Ryan B.; Elkins, Christopher J. (2003). "Rapid in-vitro physiologic flow experimentation using rapid prototyping and particle image velocimetry" (<http://www.tulane.edu/~sbc2003/pdfdocs/0419.PDF>) (PDF). *2003 Summer Bioengineering Conference*: 419. Retrieved 2007-10-10.

37. D. J. Stevenson; T. K. Lake; B. Agate; V. Gárcés-Chávez; K. Dholakia; F. Gunn-Moore (2006-10-16). "Optically guided neuronal growth at near infrared wavelengths" (<http://www.opticsinfobase.org/viewmedia.cfm?uri=oe-14-21-9786&seq=0>). *Optics Express*. **14** (21): 9786–93. Bibcode:2006OExpr..14.9786S (<https://ui.adsabs.harvard.edu/abs/2006OExpr..14.9786S>). doi:10.1364/OE.14.009786 (<https://doi.org/10.1364%2FOE.14.009786>). PMC 2869025 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2869025>). PMID 19529370 (<https://pubmed.ncbi.nlm.nih.gov/19529370>).
38. Coxworth, Ben (April 21, 2011). "Laser igniters could spell the end for the humble spark plug" (<http://www.gizmag.com/laser-ignition-spark-plug-alternative/18469/>). *Gizmag*. Retrieved March 30, 2012.
39. Pavel, Nicolaie; et al. (2011). "Composite, all-ceramics, high-peak power Nd:YAG/Cr⁴⁺:YAG monolithic micro-laser with multiple-beam output for engine ignition" (<https://doi.org/10.1364%2FOE.19.009378>). *Optics Express*. **19** (10): 9378–84. Bibcode:2011OExpr..19.9378P (<https://ui.adsabs.harvard.edu/abs/2011OExpr..19.9378P>). doi:10.1364/OE.19.009378 (<https://doi.org/10.1364%2FOE.19.009378>). PMID 21643194 (<https://pubmed.ncbi.nlm.nih.gov/21643194>).
40. "Anti-Personnel Lasers - Iraq Special Weapons" (<https://fas.org/nuke/guide/iraq/other/laser.htm>).
41. Fisher, Franklin (2003-05-14). "U.S. says Apache copters were targeted by laser weapons near Korean DMZ" (<https://www.stripes.com/news/u-s-says-apache-copters-were-targeted-by-laser-weapons-near-korean-dmz-1.9753>). *Stars and Stripes*. Retrieved 2016-12-20.
42. Lister, Tim. "North Korea's military aging but sizeable" (https://web.archive.org/web/20101126153445/http://articles.cnn.com/2010-11-24/world/north.korea.capability_1_military-cooperation-massive-military-parade-pyongyang?_s=PM:WORLD). CNN. Archived from the original (http://articles.cnn.com/2010-11-24/world/north.korea.capability_1_military-cooperation-massive-military-parade-pyongyang?_s=PM:WORLD) on 2010-11-26. Retrieved 24 December 2010.
43. "Cavity Ring Down Spectroscopy (CRDS) : Atmospheric Chemistry: HIRAC Group" (<https://hirac.leeds.ac.uk/instrumentation/crds/?adlt=strict&toWww=1&redig=94B32D323B514B6F9FFF7240AF79D5AA>). *hirac.leeds.ac.uk*. Retrieved 2023-01-04.
44. F. P. Schäfer (Ed.), *Dye Lasers* (Springer-Verlag, Berlin, 1990).
45. F. J. Duarte, *Tunable Laser Optics* (Elsevier-Academic, New York, 2003).
46. Paschotta, Rüdiger (2008-01-15). "Frequency Doubling" (<https://www.spiedigitallibrary.org/ebooks/FG/Field-Guide-to-Lasers/110/Frequency-Doubling/10.1117/3.767474.p110?SSO=1>). *Field Guide to Lasers*. Vol. FG12. doi:10.1117/3.767474.p110 (<https://doi.org/10.1117%2F3.767474.p110>). ISBN 9780819478269.
47. Komatsu, R.; Sugawara, T.; Sassa, K.; Sarukura, N.; Liu, Z.; Izumida, S.; Segawa, Y.; Uda, S.; Fukuda, T.; Yamanouchi, K. (1997-06-30). "Growth and ultraviolet application of Li₂B₄O₇ crystals: Generation of the fourth and fifth harmonics of Nd:Y₃Al₅O₁₂ lasers" (<https://aip.scitation.org/doi/10.1063/1.119210>). *Applied Physics Letters*. **70** (26): 3492–3494. Bibcode:1997ApPhL..70.3492K (<https://ui.adsabs.harvard.edu/abs/1997ApPhL..70.3492K>). doi:10.1063/1.119210 (<https://doi.org/10.1063%2F1.119210>). ISSN 0003-6951 (<https://search.worldcat.org/issn/0003-6951>).
48. "Green Lasers" (<https://www.optotronics.com/green-lasers.php>). *www.optotronics.com*. Retrieved 2021-05-05.
49. "Nd:YAG laser" (<http://www.scientificlib.com/en/Physics/Laser/NdYAGLaser.html>). *www.scientificlib.com*. Retrieved 2021-05-05.

50. Fan, T. Y.; Byer, R. L. (1987-10-01). "Continuous-wave operation of a room-temperature, diode-laser-pumped, 946-nm Nd:YAG laser" (<https://pubmed.ncbi.nlm.nih.gov/19741880/>). *Optics Letters*. **12** (10): 809–811. Bibcode:1987OptL...12..809F (<https://ui.adsabs.harvard.edu/abs/1987OptL...12..809F>). doi:10.1364/ol.12.000809 (<https://doi.org/10.1364%2Fol.12.000809>). ISSN 0146-9592 (<https://search.worldcat.org/issn/0146-9592>). PMID 19741880 (<https://pubmed.ncbi.nlm.nih.gov/19741880/>).
51. Keiderling, Tim (2013). "Chem 542 Techniques of Optical Spectroscopy in Analytical Chemistry" (<http://www2.chem.uic.edu/tak/chem52413/>). *www2.chem.uic.edu*. <http://www2.chem.uic.edu/tak/chem52413/notes3/notes3b-13sol.pdf>. Retrieved 2021-05-05.
52. Paschotta, Rüdiger. "YAG Lasers" (https://www.rp-photonics.com/yag_lasers.html). *Encyclopedia of Laser Physics and Technology*. RP Photonics. Retrieved 2018-01-16.
- Siegman, Anthony E. (1986). *Lasers* (<https://archive.org/details/lasers0000sieg>). University Science Books. ISBN 978-0-935702-11-8.
 - Koechner, Walter (1988). *Solid-State Laser Engineering* (2nd ed.). Springer-Verlag. ISBN 978-3-540-18747-9.

Retrieved from "https://en.wikipedia.org/w/index.php?title=Nd:YAG_laser&oldid=1294839838"